Information for Candidates

Numerical Answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

Notation

The following notation is used in this paper:

- The voltage waveform at node X in an AC circuit is denoted by $x(t)$, while the phasor voltage is denoted by X.
- Component and source values are generally given in Ohms, Farads, Henrys, Volts, or Amps, unless otherwise stated.
- Times are in seconds, unless otherwise stated.

1. (a) Using nodal analysis, determine the voltages at nodes X and Y in the circuit shown in Figure 1a. Then, calculate the Thévenin equivalent circuit for the entire network (including the 6 kΩ resistor) between nodes X and Y. [6]

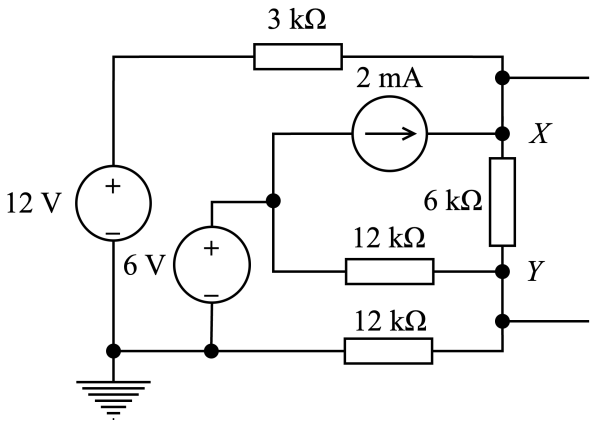


Figure 1a

Solution: At node X:

$$\frac{X-12}{3000}-\frac{2}{1000}+\frac{X-Y}{6000}=0$$

At node Y:

$$\frac{Y}{12000}+\frac{Y-X}{6000}+\frac{Y-6}{12000}=0$$

Solving the equations, we find:

$$X=15\text{ V},\quad Y=9\text{ V}$$

The Thévenin voltage is:

$$V_{\text{th}}=X-Y=6\text{ V}$$

The Thévenin resistance is:

$$R_{\text{th}}=\left(3000+\frac{1}{\frac{1}{12000}+\frac{1}{12000}}\right)\parallel 6000=9000\parallel 6000=3600\ \Omega$$

(b) For the filter circuit shown in Figure 1b, determine the transfer function $H(j\omega)=\dfrac{Y(j\omega)}{X(j\omega)}$. [8]

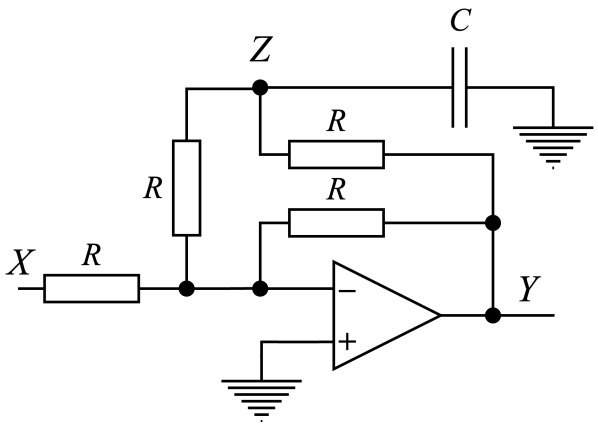


Figure 1b

Solution:
The circuit has negative feedback, so we can assume $V_+ = V_- = 0$.
Applying KCL at V_- gives:

$$\frac{0-X}{R} + \frac{0-Y}{R} + \frac{0-Z}{R} = 0$$

from which $-Z = X + Y$.
Now, applying KCL at node Z:

$$(Z-0)j\omega C + \frac{Z-Y}{R} + \frac{Z}{R} = 0$$

from which:

$$Y - Z(2 + j\omega RC) = 0$$

Substituting $-Z = X + Y$ gives:

$$Y + (X+Y)(2 + j\omega RC) = 0$$

from which:

$$H(j\omega) = \frac{Y}{X} = -\frac{2 + j\omega RC}{3 + j\omega RC}$$

(c) For the circuit shown in Figure 1c, assume the diode has a forward voltage drop of 0.7 V. Determine the value of U at which the diode switches between operating regions. Write an expression for the output voltage Y in terms of U for both operating modes. [6]

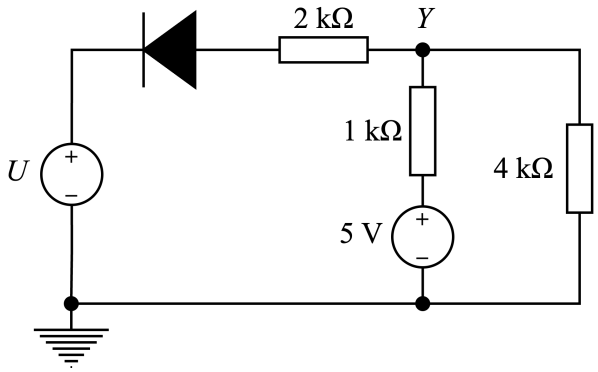


Figure 1c

Solution:

When the diode is off, $I_D = 0$, and when the diode is on, $V_D = 0.7\,\text{V}$. The diode switches operating regions at the point where both conditions are simultaneously satisfied: $I_D = 0$ and $V_D = 0.7\,\text{V}$.

When the diode is off — including at the switching point where $I_D = 0$ — the output voltage is determined by the voltage divider:

$$Y = \frac{4}{5} \times 5 = 4\,\text{V}$$

At the switching point, where $V_D = 0.7\,\text{V}$ and $I_D = 0$, using $V_D = Y - U = 0.7\,\text{V}$, we find:

$$U = Y - 0.7 = 4 - 0.7 = 3.3\,\text{V}$$

which is the switching point.

When $U < 3.3\,\text{V}$, the diode is on, and the voltage at the node between the diode and the $2\,\text{k}\Omega$ resistor is $U + 0.7$. Applying KCL at node Y:

$$\frac{Y - (U + 0.7)}{2\,\text{k}\Omega} + \frac{Y - 5}{1\,\text{k}\Omega} + \frac{Y}{4\,\text{k}\Omega} = 0$$

$$Y = \frac{2U + 21.4}{7}$$

Operating regions:

- For $U < 3.3\,\text{V}$, the diode is **on**, and $Y = \dfrac{2U + 21.4}{7}$
- For $U \geq 3.3\,\text{V}$, the diode is **off**, and $Y = 4\,\text{V}$ (constant)

(d) State the ideal properties of an amplifier and the limitations in achieving these with a single-stage common emitter configuration. [8]

Solution:

- The amplifier should be linear, i.e. the output should have the same shape as the input (same ratio and relative phase of all frequencies). This is achieved by operating only with small signal input voltages.

- The gain should be large and easy to set during design time. This is achieved using an active load.
- The input impedance should be as high as possible so that the amplifier does not load the source. This is achieved with a combination of the bias network and the high r_{be} of the transistor.
- The output impedance should ideally be low, but this is hard to achieve with a single-stage configuration with high gain.

(e) Calculate the small signal voltage gain of the amplifier shown in Figure 1e. You may assume that the the capacitor C_E is a short circuit at signal frequencies. [7]

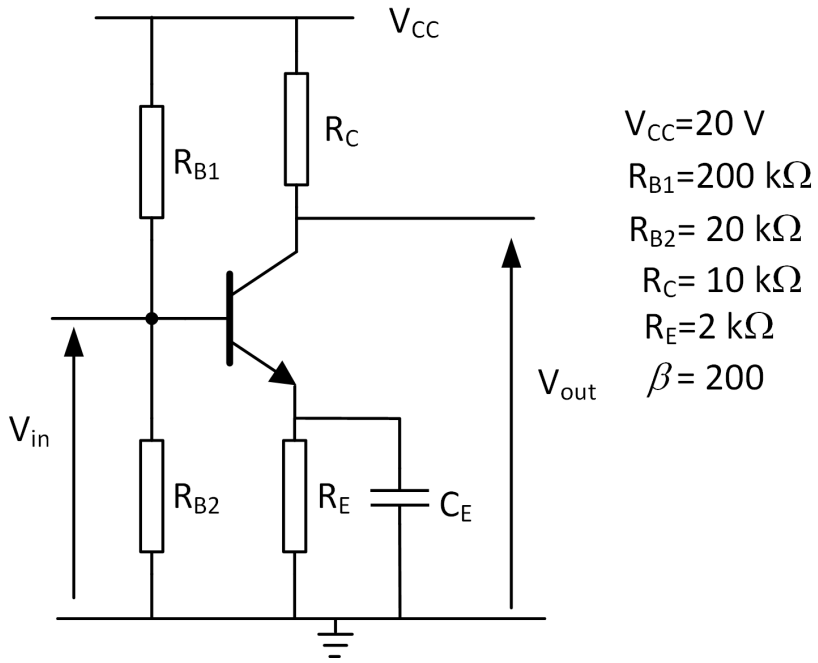
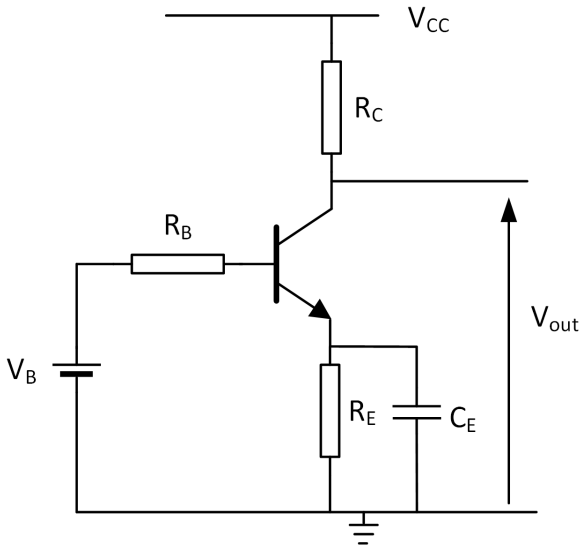


Figure 1e

Solution:

First find the bias point. This is most easily done by transforming R_{B1} and R_{B2} into a Thevenin equivalent network. This would give a Thevenin resistance of $R_B = R_{B1}//R_{B2}$ and a voltage source of $V_B = V_{CC}.R_{B2}/(R_{B1} + R_{B2})$, i.e.:



$R_B = 18.18k$ and $V_B = 1.82$
$V_B = I_E R_E + V_{BE} + I_B R_B$
This gives a bias emitter current of 0.536 mA and hence a collector current of 0.533 mA. This gives a transconductance of $g_m = 0.5333/25 = 21.3\,mS$
Which gives a small signal voltage gain of -21.3×10 = −213

(f) Calculate the quiescent output voltage in the circuit of Figure 1f. [5]

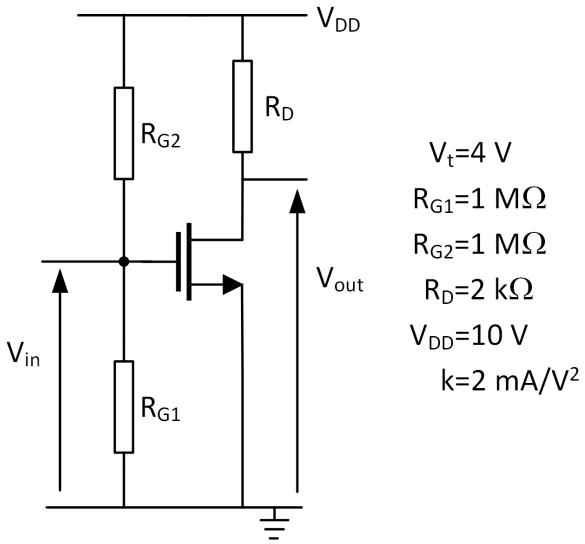


Figure 1f

Solution:
V_{GS} is 5 V
Assume the MOSFET is in the saturation mode, and hence $I_D = k(V_{GS} - V_t)^2$
This gives $I_D = 2\,mA$
Which gives $V_{out} = 6\,V$ and a check shows that $V_{DS} > V_{GS} - V_t$.

2. For the circuit shown in Figure 2, answer the following:

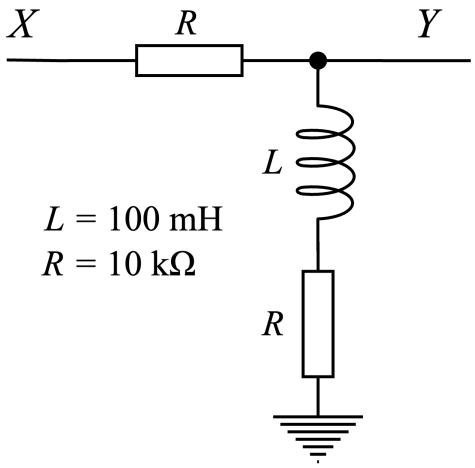


Figure 2

(a) Find the transfer function $H(j\omega) = \dfrac{Y(j\omega)}{X(j\omega)}$. [6]

Solution:

The output voltage Y is taken across a resistor R and inductor L in series. This branch is in series with another resistor R, and the input voltage is X.

The impedance of the output branch is:

$$Z_{\text{out}} = R + j\omega L$$

The total impedance in the circuit is:

$$Z_{\text{total}} = R + (R + j\omega L) = 2R + j\omega L$$

Using the voltage divider rule, the transfer function is:

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{Z_{\text{out}}}{Z_{\text{total}}} = \frac{R + j\omega L}{2R + j\omega L}$$

Substituting the component values: $R = 10\,\text{k}\Omega = 10^4\ \Omega$, $L = 100\,\text{mH} = 0.1$ H:

$$H(j\omega) = \frac{10^4 + j\omega \cdot 0.1}{2 \cdot 10^4 + j\omega \cdot 0.1}$$

(b) Sketch a straight-line (asymptotic) approximation of the frequency response between 100 Hz and 1 MHz. Both the magnitude and phase plots are required. Each plot carries half of the total marks. [8]

Solution:

We are given the transfer function:

$$H(j\omega) = \frac{R + j\omega L}{2R + j\omega L}$$

To construct the Bode plot, we analyse the magnitude and phase separately.

Magnitude Asymptotes:

At low frequencies ($\omega \ll \frac{R}{L}$):

$$|H(j\omega)| \approx \frac{R}{2R} = \frac{1}{2} \quad \Rightarrow \quad 20\log_{10}|H(j\omega)| = 20\log_{10}\left(\frac{1}{2}\right) = -6 \text{ dB}$$

At high frequencies ($\omega \gg \frac{R}{L}$):

$$|H(j\omega)| \approx \frac{j\omega L}{j\omega L} = 1 \quad \Rightarrow \quad 20\log_{10}|H(j\omega)| = 0 \text{ dB}$$

Corner Frequencies:

The two corner frequencies occur when the imaginary and real parts of the numerator and denominator are equal in magnitude:

$$\omega_1 = \frac{R}{L}, \quad \omega_2 = \frac{2R}{L}$$

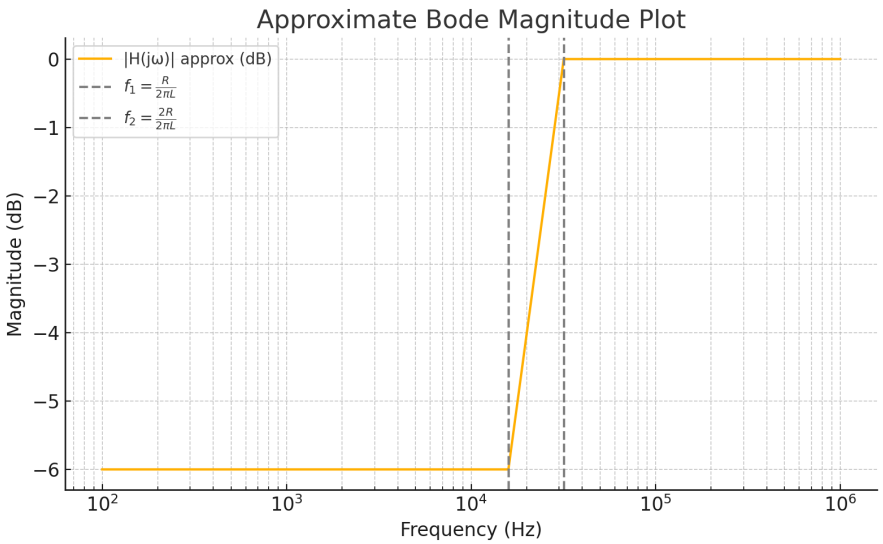
Substituting component values: $R = 10^4\ \Omega, L = 0.1$ H:

$$\omega_1 = \frac{10^4}{0.1} = 10^5 \text{ rad/s}, \quad \omega_2 = \frac{2 \cdot 10^4}{0.1} = 2 \cdot 10^5 \text{ rad/s}$$

$$f_1 = \frac{10^5}{2\pi} \approx 15.9 \text{ kHz}, \quad f_2 = \frac{2 \cdot 10^5}{2\pi} \approx 31.8 \text{ kHz}$$

Magnitude Plot Summary:

- Flat at -6 dB for $\omega < \omega_1$
- Asymptotic transition from -6 dB to 0 dB between ω_1 and ω_2
- Flat at 0 dB for $\omega > \omega_2$



Straight-line Bode Magnitude Plot for $H(j\omega)$

Phase Plot:
We are given the transfer function:

$$H(j\omega) = \frac{R + j\omega L}{2R + j\omega L}$$

The phase of the transfer function is:

$$\angle H(j\omega) = \tan^{-1}\left(\frac{\omega L}{R}\right) - \tan^{-1}\left(\frac{\omega L}{2R}\right)$$

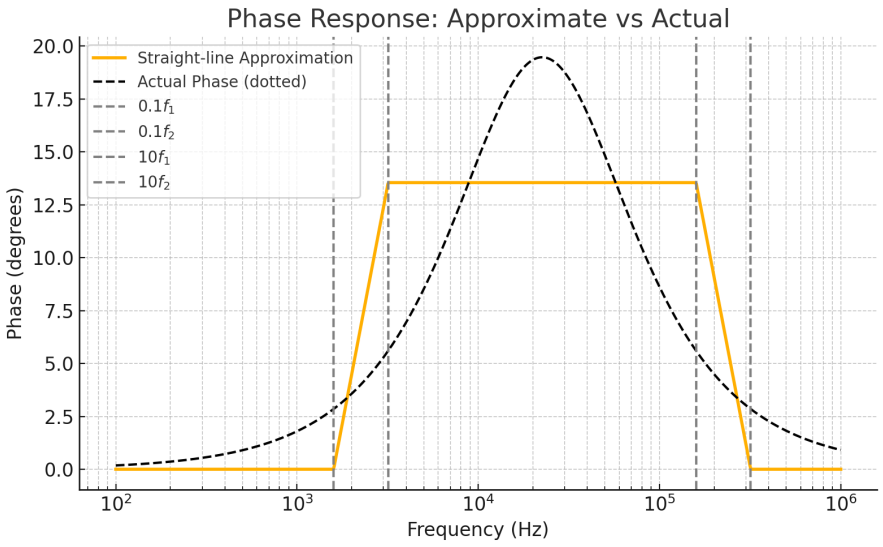
We construct a straight-line approximation for the phase plot as follows: At low frequencies ($\omega \ll \frac{R}{L}$), both arctangents are small, so the phase is approximately $0°$. At high frequencies ($\omega \gg \frac{2R}{L}$), both arctangents approach $90°$, so their difference again approaches $0°$.
There are two corner frequencies:

$$\omega_1 = \frac{R}{L}, \quad \omega_2 = \frac{2R}{L}$$

The four phase transitions occur at:

$$\omega = 0.1\omega_1, \quad \omega = 0.1\omega_2, \quad \omega = 10\omega_1, \quad \omega = 10\omega_2$$

The slope changes in the following way: Between $0.1\omega_1$ and $10\omega_1$, the phase increases linearly at $+45°$ per decade (due to the first corner frequency). Between $0.1\omega_2$ and $10\omega_2$, the phase decreases linearly at $-45°$ per decade (due to the second corner frequency). Outside these transition regions, the phase is approximated as flat (i.e., $0°$).



Comparison of Actual Phase Response and Straight-line Approximation for $H(j\omega)$

(c) Given that the input phasor is $X = 5 - 5j$, and assuming an angular frequency of $\omega = 2\pi \times 10^3$ rad/s, determine the output phasor Y. [6]

Solution:
Given the input phasor $X = 5 - 5j$, and the angular frequency $\omega = 2000\pi$ rad/s, we use the transfer function:

$$H(j\omega) = \frac{R + j\omega L}{2R + j\omega L}$$

Substitute the values $R = 10^4\ \Omega$, $L = 0.1$ H, and $\omega = 2000\pi$:

$$H(j\omega) = \frac{10^4 + j(2000\pi)(0.1)}{2 \cdot 10^4 + j(2000\pi)(0.1)} = \frac{10^4 + j628.32}{20000 + j628.32}$$

To find the output phasor:

$$Y = H(j\omega) \cdot X = \left(\frac{10000 + j628.32}{20000 + j628.32}\right)(5 - 5j)$$

Now compute the complex division and multiplication:
Let

$$H(j\omega) \approx \frac{10000 + j628.32}{20000 + j628.32} = \frac{10000 + j628.32}{20000 + j628.32}$$

Multiply this with $X = 5 - 5j$ to get:

$$Y \approx H(j\omega) \cdot (5 - 5j)$$

$$Y \approx 2.58 - 2.42j \qquad \text{or} \qquad Y \approx 3.54\angle{-43.2}^\circ$$

(d) Obtain an expression for $y(t)$ when the input signal $x(t)$ is as shown in Figure 2d, with: [10]

$$x(t) = \begin{cases} 5\sin(400000\pi t), & t < 0 \\ 10\cos(200000\pi t), & t \geq 0 \end{cases}$$

The expression for $y(t)$ should include the steady-state response before and after the switching at $t = 0$, as well as any transient response generated by the switching.

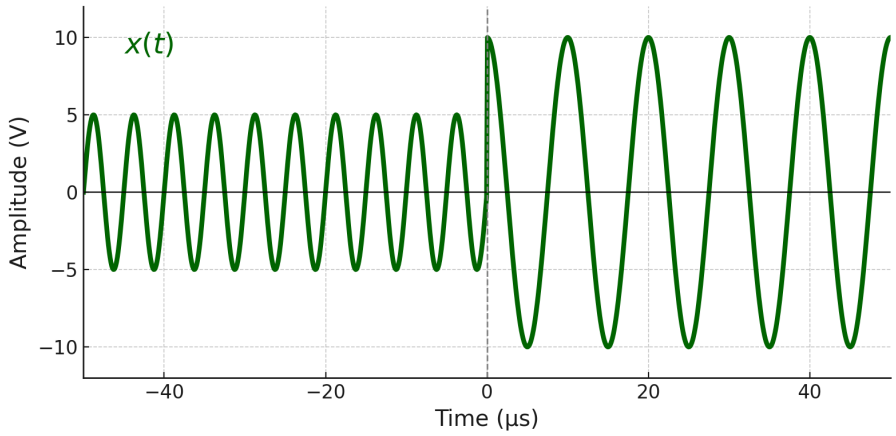


Figure 2d

Solution:
The input signal is:

$$x(t) = \begin{cases} 5\sin(400000\pi t), & t < 0 \\ 10\cos(200000\pi t), & t \geq 0 \end{cases}$$

We use the transfer function:

$$H(j\omega) = \frac{R + j\omega L}{2R + j\omega L}$$

with component values $R = 10^4\,\Omega$ and $L = 0.1\,\text{H}$.
The circuit acts as a high-pass filter with time constant:

$$\tau = \frac{L}{2R} = \frac{0.1}{2 \cdot 10^4} = 5\,\mu\text{s}$$

We evaluate the frequency response at the two frequencies:
For $t < 0$: The angular frequency is $\omega_1 = 400000\pi\,\text{rad/s}$ (200 kHz), and the input is a sine wave.

$$|H(j\omega_1)| \approx 0.9907, \quad \angle H(j\omega_1) \approx 4.5^\circ \Rightarrow y(t) = 4.95\sin(400000\pi t + 4.5^\circ)$$

For $t \geq 0$: The angular frequency is $\omega_2 = 200000\pi\,\text{rad/s}$ (100 kHz), and the input is a cosine wave.

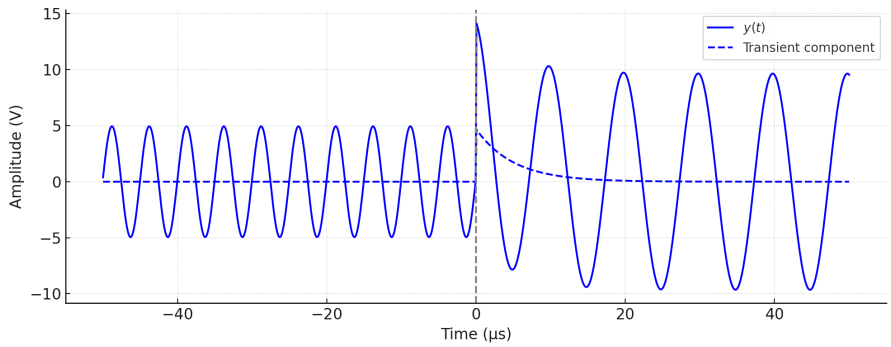
$$|H(j\omega_2)| \approx 0.9649, \quad \angle H(j\omega_2) \approx 8.6^\circ \Rightarrow y(t) = 9.65\cos(200000\pi t + 8.6^\circ)$$

Transient response: At $t = 0$, the steady-state output jumps from 4.95 V to 9.65 V, creating a transient:

$$\Delta y = 9.65 - 4.95 = 4.7\,\text{V} \Rightarrow y_{\text{transient}}(t) = 4.7e^{-t/5\mu\text{s}}, \quad t \geq 0$$

Final complete response:

$$y(t) = \begin{cases} 4.95\sin(400000\pi t + 4.5^\circ), & t < 0 \\ 9.65\cos(200000\pi t + 8.6^\circ) + 4.7e^{-t/5\mu\text{s}}, & t \geq 0 \end{cases}$$



3. This question concerns the differential amplifier show in Figure 3. All transistors are matched.

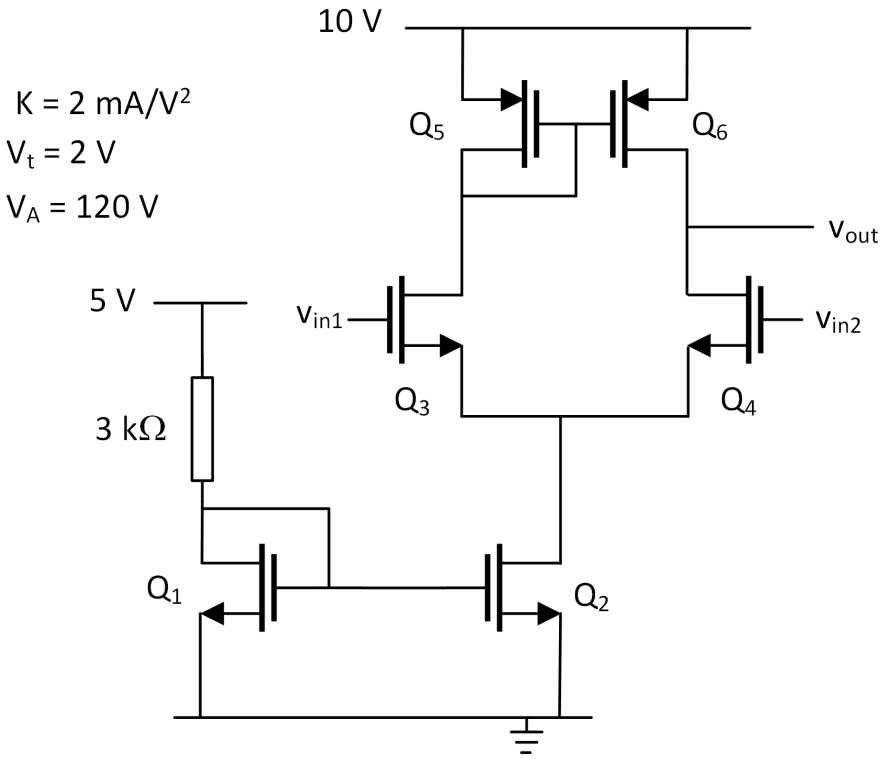


Figure 3

(a) Define and explain the terms differential-mode gain, common-mode gain, and common mode rejection ratio (CMRR). [7]

Solution:

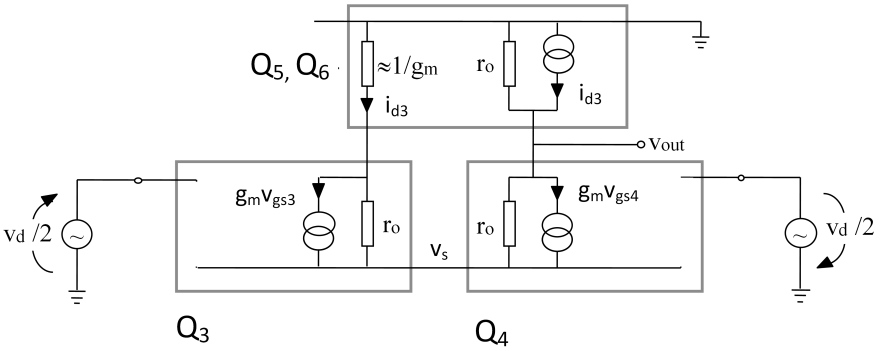
- The output voltage of a differential amplifier is defined by $v_{out} = A_d(v_+ - v_-) + A_{cm}(v_+ + v_-)/2$, where A_d is the differential model gain and A_{cm} is the common mode gain.
- The differential amplifier should have a high A_d so that the difference between the input signals is amplified.
- The differential amplifier should have a very low (ideally zero) A_{cm}, so that changes in average input voltage have little to no effect on the output.
- CMRR=A_d/A_{cm}

(b) Calculate the tail current (the drain current of Q_2) in the circuit of Figure 3. Ignore channel length modulation in your calculation. [5]

Solution:
$I_{D_{Q_1}} = \frac{5-V_{GS}}{3} = k(V_{GS} - V_t)^2$
$I_{D_{Q_1}} = \frac{5-V_{GS}}{3} = 2(V_{GS} - 2)^2$
$\therefore V_{GS} = 2.63$ or 1.20. Only the value that exceeds V_t is physical, hence $V_{GS} = 2.63V$. This sets the tail current to 0.79 mA.

(c) Draw a small signal model of the differential amplifier under differential mode excitation (i.e. when $v_{in1} = -v_{in2} = v_d/2$). [4]

Solution:
The small signal model in differential mode assumes output impedance of the low side current mirror is infinite and results in:



(d) Calculate the small signal voltage gain and state any assumptions. [5]

Solution:
The current that flows in each side of the differential pair is equal under quiescent conditions, and hence the drain current in Q_3 and Q_4 is 0.395 mA.
This gives a value of $g_m = 2\sqrt{kI_D} = 1.78\,mS$ for all 4 transistors in the model.
The output resistance of each transistor is $V_A/I_D = 304k\Omega$.
The small signal voltage gain (which can be derived from the SSEM) is $g_m.r_o/2$.
This assumes that the small-signal source voltage of Q_3 and Q_4 is small compared to the output voltage. This gain evaluates to 270.

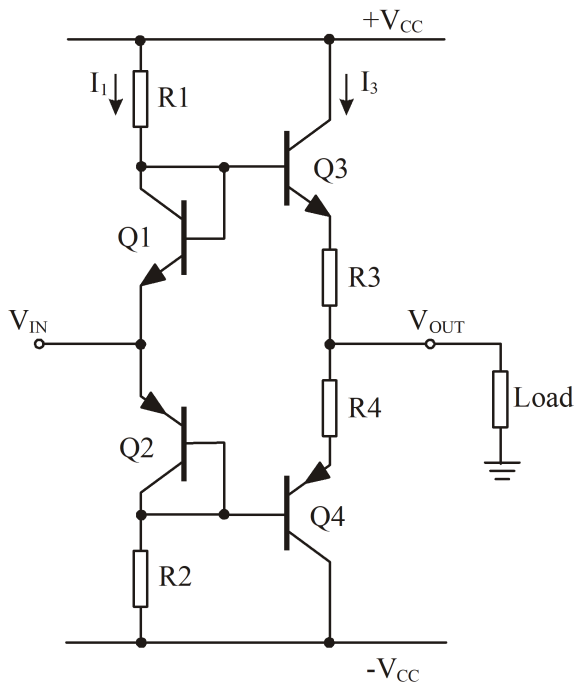
(e) Calculate the minimum common-mode voltage on the inputs so that the amplifier remains within its common mode operating range. [3]

Solution:
We require transistor Q_2 to stay in saturation for the differential amplifier to operate in its CMVR.
V_{GS} on each of Q_3 and Q_4 needs to be calculated. We know each device conducts 0.395 mA.
By rearrangement of $I_D = k(V_{GS} - V_t)^2$, this requires a gate-source voltage of 2.44 V.
For Q_2 to remain in saturation, $V_{DS} \geq V_{GS} - V_t$, and so $V_{DS_{Q2}} \geq 2.63 - 2$=0.63 V. This means the minimum input voltage is 0.63+2.44=3.07 V

(f) It is proposed to add a BJT based class AB stage to the output of the differential amplifier. Draw a class AB stage and explain: [6]

- How cross-over distortion is almost eliminated.
- How shoot-though current between the power supply rail and ground is controlled.

Solution:
A practical class AB circuit is shown below.



- Cross over distortion is almost eliminated by concurrently adding and subtracting one base-emitter voltage from V_{in} with transistors Q_1 and Q_2, which keeps Q_3 and Q_4 biased to be just conducting.
- Shoot through currents are limited under zero input voltage through the use of current limiting resistors, R_3 and R_4 as voltage dropped across these reduces the base-emitter voltage in the output devices. As the input voltage increases (> 0), increased current in R_3 reduces the output voltage slightly, reducing the base-emitter voltage of Q_4 and turning it off. The opposite is true for negative input voltages, and hence shoot through current is minimal for large (positive and negative) input voltages.